Parameter Estimation: 10 Hard Problems

MLE Techniques · German Tank/MVUE · Bias/Var/MSE · MoM Failure Modes · Capstone

Same philosophy as your other hard sets: every problem here fuses two or more ideas from the base 20-problem sheet, and several are direct escalations — H3 and H9 both reuse the bias-variance-optimization machinery from H8 across two completely different parameters, H10 forces a three-way MSE ranking you've never had to compute at once before, and H4/H8 are deliberate "the formula breaks" traps in the same spirit as the base sheet's C2. Every option set has exactly one correct numeric answer and three answers built from a specific, common error.

I. MLE Techniques Beyond the Standard Cases

Problem H1 — Hypergeometric MLE via the discrete ratio test

A lake's fish population N is unknown. You tag $n_1 = 20$ fish and release them. Later you catch $n_2 = 15$ fish, of which $m = 5$ are tagged. Using the hypergeometric likelihood $L(N) = \frac{\binom{n_1}{m}\binom{N-n_1}{n_2-m}}{\binom{N}{n_2}}$ and the discrete MLE technique (find N where $L(N)/L(N-1)$ crosses 1), find the MLE of N.

Options: (A) 55 (B) 60 (C) 65 (D) 75

Solution. For discrete parameters without a calculus optimum, use the ratio test: $L(N)$ is increasing while $L(N)/L(N-1) > 1$ and decreasing afterward, so the MLE is the integer where this ratio first drops below 1. Working through the ratio (a classical result) gives the closed form:

$$\hat{N}_{\text{MLE}} = \left\lfloor \frac{n_1 n_2}{m} \right\rfloor = \left\lfloor \frac{20 \times 15}{5} \right\rfloor = \lfloor 60 \rfloor = \boxed{60}.$$

Trap analysis: 55 and 65 are plausible-looking round numbers near the answer with no actual derivation behind them. 75 comes from computing $n_1 + n_2 \times m$ or a similarly mangled formula instead of $n_1 n_2/m$ — a common error when rushing the ratio-test algebra under time pressure.

Correct: (B) 60

▶ *Speed insight:* This is the same formula behind the Lincoln–Petersen index from Notebook 3's capture-recapture methods — recognizing "tag, release, recapture, count overlaps" should immediately trigger $\hat{N} = n_1 n_2/m$ without re-deriving the ratio test from scratch every time.

Problem H2 — Two-parameter boundary MLE: both endpoints unknown

A rival desk's ticket numbers run consecutively from unknown start A to unknown end B. You observe 6 tickets: $114, 137, 108, 129, 122, 119$. Find the MLE of the total number of tickets issued, $B - A + 1$.

Options: (A) 24 (B) 29 (C) 30 (D) 36

Solution. For discrete Uniform$\{A, \ldots, B\}$ with both endpoints unknown, $L(A, B) = (B - A + 1)^{-k}$ for $A \le$ all data $\le B$ — maximized by making the interval *as tight as possible* around the observed data:

$$\hat{A} = \min(\text{data}) = 108, \qquad \hat{B} = \max(\text{data}) = 137.$$

$$\widehat{B - A} + 1 = 137 - 108 + 1 = \boxed{30}.$$

Trap analysis: 29 forgets the "+1" (computing max − min instead of the count of integers from min to max inclusive). 24 uses the *range of the middle four values* after incorrectly discarding the min and max as "outliers." 36 adds an arbitrary margin, as if hedging against the boundary bias (a reasonable instinct, but not what MLE actually computes).

Correct: (C) 30

▶ *Speed insight:* Two unknown boundary parameters just means two separate applications of the same "shrink the interval until it just barely contains the data" logic from the single-boundary case — min for the lower endpoint, max for the upper, independently.

II. The German Tank Problem, Revisited

Problem H3 — Is the Method of Moments estimator *also* unbiased?

For the German Tank setup, you know $\hat{N}_{\text{MVUE}} = m + \frac{m}{k} - 1$ is unbiased. Is the Method of Moments estimator $\hat{N}_{\text{MoM}} = 2\bar{X} - 1$ *also* unbiased for N? If so, find $\text{Var}(\hat{N}_{\text{MoM}})$ at $N = 100, k = 5$ and compare to $\text{Var}(\hat{N}_{\text{MVUE}}) \approx 274.1$.

Options: (A) No, biased; Var= 274.1 (B) Yes, unbiased; Var≈ 274.1 (C) Yes, unbiased; Var≈ 639.7 (D) Yes, unbiased; Var≈ 116.2

Solution. The sample mean $\bar{X}$ from simple random sampling *without replacement* is always unbiased for the population mean, regardless of k: $\mathbb{E}[\bar{X}] = \frac{N+1}{2} \implies \mathbb{E}[\hat{N}_{\text{MoM}}] = N$.

Yes, it's unbiased too! Using $\text{Var}(\bar{X}) = \frac{(N+1)(N-k)}{12k}$ (finite-population sampling variance):

$$\text{Var}(\hat{N}_{\text{MoM}}) = 4\,\text{Var}(\bar{X}) = \frac{(N+1)(N-k)}{3k}$$

$$= \frac{101(95)}{15} \approx \boxed{639.7}.$$

So MoM is unbiased *but* has $639.7/274.1 \approx 2.33\times$ the variance of the MVUE — confirming the MVUE is strictly more efficient despite both being unbiased.

Trap analysis: 274.1 (options A, B) wrongly assumes MoM must match the MVUE's variance since both are unbiased — unbiasedness says nothing about variance. 116.2 is $\text{Var}(\hat{N}_{\text{MVUE}})$ at a different k, a mismatched-parameter trap.

Correct: (C) Yes, unbiased; Var≈639.7

▶ *Speed insight:* In general, $\text{Var}(\hat{N}_{\text{MoM}}) = \text{Var}(\hat{N}_{\text{MVUE}}) \times \frac{k+2}{3}$ — memorize this ratio and you can compare the two estimators' efficiency instantly for any k, without recomputing either variance from scratch. At $k = 1$ the ratio is exactly 1 (the two estimators coincide!), and it grows without bound as k increases.

Problem H4 — Inverse design: how many tanks must you capture?

Suspecting $N = 500$, you want the MVUE's standard deviation to be *under* 10 tanks. Using $\text{Var}(\hat{N}) = \frac{(N-k)(N+1)}{k(k+2)}$, find the minimum integer sample size k.

Options: (A) 40 (B) 44 (C) 47 (D) 52

Solution. Require $\frac{(500-k)(501)}{k(k+2)} < 100$. Clearing denominators: $501(500-k) < 100k(k+2)$, i.e.

$$100k^2 + 701k - 250500 > 0.$$

Solving the quadratic: $k > \frac{-701 + \sqrt{701^2 + 4(100)(250500)}}{200} \approx 46.67$. Since k must be an integer, check $k = 46$ (sd ≈ 10.15, still too high) and $k = 47$ (sd$\approx 9.93 < 10$):

$$\hat{k}_{\min} = \boxed{47}.$$

Trap analysis: 40 and 44 round the unrounded root (46.67) incorrectly or solve a mis-expanded quadratic. 52 over-corrects,

perhaps from requiring Var < 10 instead of sd < 10 (i.e. squaring in the wrong direction).

Correct: (C) 47

▶ *Speed insight:* Always double check an inverse/"solve for k" answer by plugging the two neighboring integers back into the *original* inequality (not just trusting the quadratic root) — rounding direction errors are the single most common mistake on inverse-design questions under time pressure.

III. Bias–Variance Optimization, in Depth

Problem H5 — Invariance fails for unbiasedness: a nonlinear plug-in trap

$\hat{N}_{\text{MVUE}} = 60.2$ is unbiased for $N = 100$ (base sheet, $k = 5$, $\text{Var}(\hat{N}) \approx 274.1$). A trader needs $M = N/10$ (months of production, known rate 10/month) and then wants M^2. Is the plug-in $\hat{M}^2 = (\hat{N}/10)^2$ unbiased for $M^2 = 100$?

Options: (A) Yes, always (B) No, biased downward (C) No, biased upward, ≈ 102.7 (D) No, biased upward, ≈ 127.4

Solution. $\hat{M} = \hat{N}/10$ *is* unbiased for M (linear transforms preserve unbiasedness). But squaring is **nonlinear**:

$$\mathbb{E}[\hat{M}^2] = \text{Var}(\hat{M}) + (\mathbb{E}[\hat{M}])^2 = \frac{\text{Var}(\hat{N})}{100} + M^2$$

$$= \frac{274.1}{100} + 100 \approx \boxed{102.7}.$$

Biased *upward* by ≈ 2.7 — consistent with $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ always (Jensen's inequality for the convex function x^2).

Trap analysis: "Yes, always" is exactly MLE's invariance property misapplied to unbiasedness — MLE invariance holds for *any* function g, but unbiasedness only survives *linear* transforms. 127.4 forgets to divide $\text{Var}(\hat{N})$ by $10^2 = 100$ before adding. "Biased downward" has the Jensen's-inequality direction backwards — convex functions of unbiased estimators are biased *upward* on average, concave functions (like $\sqrt{\cdot}$) biased *downward*.

Correct: (C) No, biased upward, ≈102.7

Problem H6 — Solving for the optimal shrinkage factor

Generalize the base sheet's shrinkage estimator: $\tilde{p}_\alpha = \alpha\hat{p} + (1-\alpha)(0.5)$ for $\hat{p} = k/n$, $n = 10$, true $p = 0.3$. Find the $\alpha \in [0, 1]$ that *minimizes* MSE, and the resulting minimum MSE.

Options: (A) $\alpha^* = 1$, $\text{MSE}^* = 0.0210$ (B) $\alpha^* \approx 0.656$, $\text{MSE}^* \approx 0.0138$ (C) $\alpha^* \approx 0.344$, $\text{MSE}^* \approx 0.0138$ (D) $\alpha^* \approx 0.656$, $\text{MSE}^* \approx 0.0157$

Solution. Let $A = (0.5 - p)^2 = 0.04$ (squared bias of full shrinkage) and $B = \dfrac{p(1-p)}{n} = 0.021$ (MLE's variance). Then $\text{Bias}(\alpha) = (1-\alpha)(0.5-p)$, $\text{Var}(\alpha) = \alpha^2 B$, and

$$\text{MSE}(\alpha) = (1-\alpha)^2 A + \alpha^2 B.$$

Differentiating and solving: $\alpha^* = \dfrac{A}{A+B} = \dfrac{0.04}{0.061} \approx \boxed{0.656}$, and substituting back gives the clean product-over-sum form:

$$\text{MSE}^* = \frac{AB}{A+B} = \frac{(0.04)(0.021)}{0.061} \approx \boxed{0.0138}.$$

Trap analysis: $\alpha^* = 1, \text{MSE} = 0.0210$ is just the pure MLE (no shrinkage at all) — the *worst* point in this family, not the best; a student who forgot to actually differentiate would default here. $\alpha \approx 0.344 = 1 - 0.656$ swaps α and $(1-\alpha)$, a sign-convention mix-up. $\text{MSE} \approx 0.0157$ is the *fixed* Laplace-smoothing estimator's MSE from the base sheet (a specific, non-optimal member of this family) — a tempting but wrong number to recognize and misapply here.

Correct: (B) $\alpha^* \approx 0.656$, $\text{MSE}^* \approx 0.0138$

▶ *Speed insight:* The pattern $\text{MSE}^* = \dfrac{AB}{A+B}$ (a "harmonic mean"-flavored formula) recurs constantly in optimal bias-variance trade-offs — recognize it and you can skip the differentiation entirely next time and jump straight to the answer once you've identified A (squared bias at full shrinkage) and B (variance of the unbiased estimator).

Problem H7 — Unbiased but wasteful: single observation vs. the full sample

For Pois(λ=5), compare using just the *first* observation X_1 to using the full sample mean $\bar{X}$ from $n = 10$ i.i.d. observations, as estimators of λ. Both are unbiased. Find the ratio $\text{MSE}(X_1)/\text{MSE}(\bar{X})$.

Options: (A) 1 (B) 5 (C) $\sqrt{10} \approx 3.16$ (D) 10

Solution. Both unbiased ($\mathbb{E}[X_1] = \mathbb{E}[\bar{X}] = \lambda$), so MSE = Var for each.

$$\text{Var}(X_1) = \lambda = 5 \text{ (Poisson property)},$$

$$\text{Var}(\bar{X}) = \frac{\lambda}{n} = \frac{5}{10} = 0.5.$$

$$\frac{\text{MSE}(X_1)}{\text{MSE}(\bar{X})} = \frac{5}{0.5} = \boxed{10}.$$

Trap analysis: 1 assumes "both unbiased $\implies$ equally good," ignoring variance entirely — the exact same trap as H3. 5 forgets to divide $\text{Var}(X_1)$ by anything, essentially reporting $\text{Var}(X_1)$ alone. $\sqrt{10}$ confuses the MSE ratio with the standard-error ratio (which does scale as $\sqrt{n}$, not n).

Correct: (D) 10

▶ *Speed insight:* For i.i.d. unbiased estimators, MSE (= variance here) scales as $1/n$ while standard deviation scales as $1/\sqrt{n}$ — always double check which one a question is actually asking for before reaching for $\sqrt{n}$.

IV. When the Method of Moments Structurally Breaks

Problem H8 — Overdispersion: a guaranteed MoM failure, not just an outlier accident

Weekly large-order counts are modeled as Binomial(n, p), both unknown. Historical data: sample mean $\bar{x} = 3$, sample variance $s^2 = 5$ (variance *exceeds* the mean). Applying $\hat{p} = 1 - s^2/\bar{x}$, $\hat{n} = \bar{x}/\hat{p}$, what goes wrong, and why is this failure *guaranteed* rather than a fluke?

Options: (A) $\hat{p} \approx 0.2$, valid (B) $\hat{p} \approx -0.667$, impossible — structural (C) $\hat{p} \approx -0.667$, impossible — a rare fluke (D) $\hat{n}$ negative, but $\hat{p}$ fine

Solution.

$$\hat{p} = 1 - \frac{5}{3} = \boxed{-0.667} \quad \text{(impossible: } p \in [0, 1] \text{ required)}.$$

This isn't bad luck: for *any* true Binomial(n, p), $\text{Var} = np(1-p) < np = \text{Mean}$ **always** (strictly, whenever $0 < p < 1$). So *any* dataset with $s^2 \geq \bar{x}$ (variance meeting or exceeding the mean — "overdispersion") is structurally incompatible with a Binomial model, guaranteed to break this MoM formula, every single time.

Trap analysis: $\hat{p} \approx 0.2$ computes $\bar{x}/(\text{something})$ without noticing the sign of $1 - s^2/\bar{x}$. "A rare fluke" misdiagnoses a structural model-mismatch as sampling noise — the real lesson is that overdispersed count data (common in real trading/claims data)

often needs a Negative Binomial model instead, precisely *because* it structurally violates Binomial's Var<Mean constraint.

Correct: (B) $\hat{p} \approx -0.667$, impossible — structural

▶ *Speed insight:* Before trusting a 2-moment MoM result for any bounded-variance family (Binomial, hypergeometric), do a 1-second sanity check: is $s^2 < \bar{x}$? If not, stop — the model itself is wrong for this data, no amount of algebra will fix it.

Problem H9 — Shrinkage, applied back to the German Tank Problem

A trader's prior guess is $N_0 = 80$; data gives $\hat{N}_{\text{MVUE}} = 100$'s-worth of information with $\text{Var}(\hat{N}_{\text{MVUE}}) \approx 274.1$ (i.e., evaluate at true $N = 100, k = 5$ as in H3). Using $\tilde{N}_\beta = \beta \hat{N}_{\text{MVUE}} + (1-\beta)(80)$, find the MSE-optimal β and resulting minimum MSE.

Options: (A) $\beta^* = 1$, $\text{MSE}^* = 274.1$ (B) $\beta^* \approx 0.593$, $\text{MSE}^* \approx 162.7$ (C) $\beta^* \approx 0.407$, $\text{MSE}^* \approx 162.7$ (D) $\beta^* \approx 0.593$, $\text{MSE}^* \approx 219.0$

Solution. Exactly H6's machinery, now with $A = (N_0 - N)^2 = (80-100)^2 = 400$ and $B = \text{Var}(\hat{N}_{\text{MVUE}}) \approx 274.1$:

$$\beta^* = \frac{A}{A+B} = \frac{400}{674.1} \approx \boxed{0.593},$$

$$\text{MSE}^* = \frac{AB}{A+B} \approx \boxed{162.7}.$$

Even a prior guess that's 20 tanks off from the truth still *helps* — shrinking toward it beats the pure MVUE's MSE of 274.1 by over 40%.

Trap analysis: $\beta^* = 1, \text{MSE} = 274.1$ is pure MVUE (no shrinkage), the naive default for anyone who forgot to optimize. $\beta \approx 0.407 = 1 - 0.593$ is the same $\alpha \leftrightarrow (1-\alpha)$ swap trap as H6. MSE ≈ 219.0 comes from averaging A and B directly instead of using the product-over-sum formula — a very tempting but incorrect shortcut.

Correct: (B) $\beta^* \approx 0.593$, $\text{MSE}^* \approx 162.7$

▶ *Speed insight:* This is the exact same formula as H6 applied to a totally different parameter (population size instead of a probability) — once you recognize the $\frac{AB}{A+B}$ shrinkage pattern, it transfers instantly across every estimation context in this entire chapter.

Problem H10 — Capstone: rank all three estimators by MSE on fresh data

New capture data, $k = 8$: $23, 71, 45, 88, 12, 67, 39, 91$ ($N = 100$ true, for grading). Using the *general* Bias/Variance formulas (not just this one sample), rank $\hat{N}_{\text{MLE}}$, $\hat{N}_{\text{MoM}}$, $\hat{N}_{\text{MVUE}}$ by MSE, best to worst.

Options: (A) MVUE < MoM < MLE (B) MVUE < MLE < MoM (C) MLE < MVUE < MoM (D) MoM < MVUE < MLE

Solution. At $N = 100, k = 8$: $\mathbb{E}[m] = \frac{8(101)}{9} \approx 89.78$, Bias ≈ -10.22, $\text{Var}(m) \approx 91.77$:

$$\text{MSE(MLE)} = (-10.22)^2 + 91.77 \approx \boxed{196.3}.$$

$$\text{MSE(MVUE)} = \text{Var}(\hat{N}_{\text{MVUE}}) = \frac{92(101)}{8(10)}$$

$$\approx \boxed{116.2} \quad \text{(unbiased)}.$$

$$\text{MSE(MoM)} = \text{Var}(\hat{N}_{\text{MoM}}) = \frac{101(92)}{3(8)} \approx \boxed{387.2} \quad \text{(unbiased)}.$$

$$\boxed{\text{MVUE}(116.2) \ < \ \text{MLE}(196.3) \ < \ \text{MoM}(387.2)}$$

Trap analysis: The genuinely surprising result here is **MLE beating MoM**, despite MLE being biased and MoM unbiased — MoM's much larger variance (recall the $(k+2)/3 \approx 3.33\times$ penalty from H3) overwhelms its zero-bias advantage at this sample size. Option (A) is the "unbiased-leaning" intuition that MoM should beat MLE since only MLE carries bias — false here. Option (D) wrongly ranks MoM first based purely on unbiasedness, the most tempting trap on this entire sheet.

Correct: (B) MVUE < MLE < MoM

▶ *Speed insight:* This is the single most important meta-lesson of the whole chapter: **bias status alone never determines an MSE ranking.** Always compute all three numbers — you cannot shortcut this ranking from intuition, even with reasonable-sounding reasoning about which estimator "uses more information."

Calibration note: H1, H2, H7, H8 are realistic 90–150s Stage-1 questions once the technique is recognized. H4, H6, H9, and H10 are genuinely Stage-2/Finals-grade (each needs either solving a quadratic under pressure or a full optimization) — but every one is a direct escalation of base-sheet machinery, so if Sections II–III of the base 20-problem sheet are solid, none of these should feel unfamiliar, only longer.